Gestation Guardian

A Mobile-First Maternal Health Monitoring Application

Abstract

Gestation Guardian is a mobile-first maternal health monitoring application designed to assist expectant mothers in tracking vital health parameters throughout pregnancy. Built as a hybrid web application using TypeScript, Vite, and an Android WebView wrapper, the app provides real-time health tracking, risk assessment, and educational guidance within a single-page application architecture.

The application features a comprehensive dashboard that displays gestational age, trimester progression, and a visual progress tracker calculated from the user's last menstrual period (LMP). Users can log blood pressure readings with systolic and diastolic values, record daily vitals including weight, sleep hours, stress levels, body temperature, blood glucose, and urine protein levels. A dedicated fetal kick counter with a timed session interface allows mothers to monitor fetal movement patterns, while a separate contraction timer tracks the duration and intervals between contractions.

At the core of the application is a risk scoring engine that evaluates the user's current health state by combining static risk factors (maternal age, parity, prior preeclampsia history, chronic hypertension, diabetes status, family history, multiple gestation, and BMI) with dynamic signals (latest blood pressure readings, proteinuria levels, blood glucose values, reported symptoms such as severe headache, visual disturbances, right upper quadrant pain, and sudden swelling, as well as trend-based weight gain analysis). The engine produces a numerical risk score and classifies the result into a Red-Amber-Green (RAG) triage band: Low, Moderate, High, or Critical. Each band triggers a corresponding clinical action recommendation displayed on the dashboard.

The data layer uses a hybrid storage architecture with Firebase Cloud Firestore as the primary backend and browser localStorage as a synchronous fallback. The application includes conditional initialization logic that detects whether valid Firebase credentials are configured; when they are not, it gracefully degrades to offline-only operation without errors. IndexedDB persistence is enabled for Firebase to support offline caching when cloud connectivity is available.

An integrated AI chatbot provides contextual pregnancy guidance through a keyword-matching mock engine that responds to queries about nutrition, iron intake, sleep positions, fetal kick counts, blood pressure monitoring, contractions, labor signs, and nausea management. The chatbot interface features a draggable floating action button, a full-screen chat overlay with a typing indicator, and asynchronous response delivery

with simulated latency.

Additional modules include a medical history recorder, a health records hub that aggregates logged data across all vitals categories, a care guide with trimester-specific educational content, a reminder system, user authentication with profile management, and a Bluetooth device integration stub prepared for future connectivity with blood pressure cuffs and fetal dopplers via the Web Bluetooth API.

The application is compiled using Vite with a legacy polyfill plugin for broad browser compatibility, and is packaged as an Android APK through a Kotlin-based WebView activity. The user interface follows a skeuomorphic design language with responsive layouts optimized for both mobile and desktop viewports.

Keywords

Maternal health, pregnancy monitoring, preeclampsia risk scoring, RAG triage, fetal kick counter, contraction timer, blood pressure tracking, hybrid mobile application, Firebase, TypeScript, Android WebView

Technology Stack

- Frontend: TypeScript, HTML5, CSS3 (Single Page Application)
- Build Tool: Vite 8.1.4 with Legacy Polyfill Plugin
- Backend: Firebase Cloud Firestore with IndexedDB offline persistence
- Local Storage: Browser localStorage (synchronous fallback)
- Mobile: Android WebView (Kotlin, Gradle)
- Testing: Playwright
- AI Module: Keyword-matching mock engine (mock-ai.ts)